Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/18/2006
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Project Title:	Micro-Gravity Induced Changes in the Control of Muscles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02215	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	05/11/2001	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
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Flight Program:	Pre/Post Flight		
Flight Assignment:	In flight development phase (first subjects on STS-114)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roy, Serge (Boston University NeuroMuscular Research Center)		
Grant/Contract No.:	NCC9-127		
Performance Goal No.:			
Performance Goal Text:			
	The understanding of the neuromuscular basis for microgravity-induced degradation in motor function is essential for the educated design of exercise programs to counter the effects of microgravity as well as make it possible to identify and validate a ground-based model for the phenomenon so that possible countermeasures can be studied in depth on Earth before being attempted in space. The goal of this project is to understand the effects of microgravity on the control of muscles by investigating the regulation of the firing activity of motor units that make up a muscle and the coordination of different muscles responsible for the control of a joint. The following specific aims are being investigated: 1) Understand the effects of microgravity on the control of motor units, 2) Compare the effects of microgravity on two muscles that are used to varying extents in space, in order to assess the potential of muscle use to curtail microgravity-induced disturbances in control properties of motor units, 3) Understand the effects of microgravity on the relative activation patterns of a group		

Task Description:	of muscles controlling one joint, and 4) Determine the course of recovery after re-exposure to Earth gravity. Short duration (Shuttle) crewmembers will serve as subjects in two general sets of procedures performed both pre-flight (L-45, -43, and -41) and post-flight [R+0, (+1 or +2), +3, +5, +7, +14]. Intramuscular electromyographic (EMG) signals will be recorded from a hand muscle, the First Dorsal Interosseous, and a knee extensor muscle, the Vastus Medialis. Surface EMG data will be recorded from muscles surrounding the knee joint. All EMG recordings will be taken while subjects trace specified trajectories displayed on a computer screen. The intramuscular EMG data will be decomposed into motor unit firing trains using a sophisticated set of algorithms constructed with artificial intelligence knowledge-based techniques being developed in our laboratory. Particular attention will be paid to detecting modifications in the control of the firing rate and recruitment behavior of motor units. The surface EMG data will be analyzed to evaluate the effects of microgravity on the coordination of muscles controlling a joint. Particular attention will be directed at comparing the results of this study to similar data obtained by us from elderly subjects to verify whether the effects of microgravity on muscle is similar to that of aging. 1) Determine the neural component to the microgravity-induced deterioration in muscle function and strength, at both the micro and macro levels. This will assist in the design of techniques for monitoring rehabilitative interventions that target the specific alterations in muscle control. These interventions may include exercises that need to be performed in space and those that will be carried out upon return to Earth. 2) Provide the knowledge base necessary to identify a ground-based model for the effects of microgravity (for example, the process of aging). The availability of a validated model for microgravity will facilitate the development of countermeasures by enabling the testing and validation of proposed countermeasures on Earth. 3) By comparing the effects of microgravity on muscles that are used to varying degrees in space, we will evaluate if low-level activity is useful in counteracting some of the effects of microgravity. This knowledge will be important in the design of countermeasures. 4) It is expected that the tracking of the recovery process in this study will identify the distinct effects of the two main the causes for microgravity- related losses in force: atrophy-based and neurally-mediated losses, since muscle force production is expected to recover from these following different timecourses. The knowledge regarding the significance of the contribution of the two components of loss will guide the design of countermeasures since different interventions may be developed to counter the effects of atrophy versus deterioration in neural control. This study aims to answer the basic scientific question regarding the effects of microgravity on the micro and macro control of muscle. By micro control we mean the control of motor units within a muscle; by macro control we mean the control of simultaneously acting muscles. With this information, this study has the potential of providing new strategies for the design of countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The understanding of the neurological modifications that occur in microgravity should provide an accelerated time-scale model for the neurological effect on motor control due to aging. Additionally, the equipment that will be developed for studying the behavior of motor units can be used by neurologists in the clinical environment to assess the severity of diseases such as stroke, Parkinson's disease, etc., as well as the success of various interventions.
Task Progress:	We have successfully collected surface and indwelling EMG and Force data in 1 astronaut subject for three days of baseline testing prior to STS-114 launch [Launch -120, -119, and -116 days] and four days of testing following this 14-day Space Shuttle mission [Landing Day +0, +1, +3, +7, +13 days]. Data were collected from the quadriceps (thigh) and first dorsal interosseus (hand) muscles. Results from the surface EMG data from the thigh indicate that the magnitude of the changes in the RMSEMG/Force ratio for each of the %MVC contractions was greater than the day-to-day variability observed over the baseline period, indicating that these changes were likely attributed to exposure to the Space flight conditions. All thigh muscles showed a similar pattern of behavior, where the greatest increase in the ratio was observed on landing day (R+0) followed by a gradual return to baseline by the 13th post-flight day (R+13) for most contractions. There are indications from the data that the changes were due to increased co-contraction at the knee. The indwelling EMG data from needle and fine-wire sensors were processed with the precision decomposition technique to obtain individual MU firing trains to extract neural control parameters. These parameters include the cross correlation among motor unit firing rates (common drive coefficient), the cross correlation of firing rates and the force output, the average firing rate and the recruitment threshold. The observed results indicate an increased variability of firing rates and force output immediately upon return to Earth. These modifications undergo re-adaptation during the remaining post-flight period.
Bibliography Type:	Description: (Last Updated: 12/18/2006)
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Articles in Peer-reviewed Journals	De Luca CJ, Adam A, Wotiz R, Gilmore LD, Nawab SH. "Decomposition of surface EMG signals." J Neurophysiol. 2006 Sep;96(3):1646-57. PMID: 16899649 , Sep-2006